

MSA-YOLO: A Multi-Scale Attention Enhancement of YOLO for Tiny-Human Detection in Aerial Search-and-Rescue Imagery

CSE 4000: Project/Thesis

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Introduction

Introduction

- Earthquake / flood / fire → UAVs survey the site far faster than ground teams, with no risk to rescuers.
- But: a person seen from altitude $\approx$ fewer than 10 pixels, amid rubble, water, smoke from UAV.
- Human detection in disaster scenes is challenging due to **occlusion and scale**
- Thousands of frames; a human operator scanning that stream will miss people
- Need for **efficient and disaster-aware detection** models. This thesis addresses it.



Fig. 1: Aerial Drone Footage of a flood affected Region[4]

Introduction(Cont'd.)



a. Collapsed building



b. Fire



c. Flood



d. Traffic Incident

Fig-02: Representative C2A scenes — human targets are tiny and numerous in every scene type [1]

- ~47 % of annotated people in C2A are **< 10 px** — at or below a standard detector's resolving limit.
- **20–40 people per scene**, many partially buried or occluded.
- Detection maturity (R-CNN → YOLO → DETR) does **not transfer**: targets are an order of magnitude
- smaller than these detectors were tuned for.

Human Detection Example

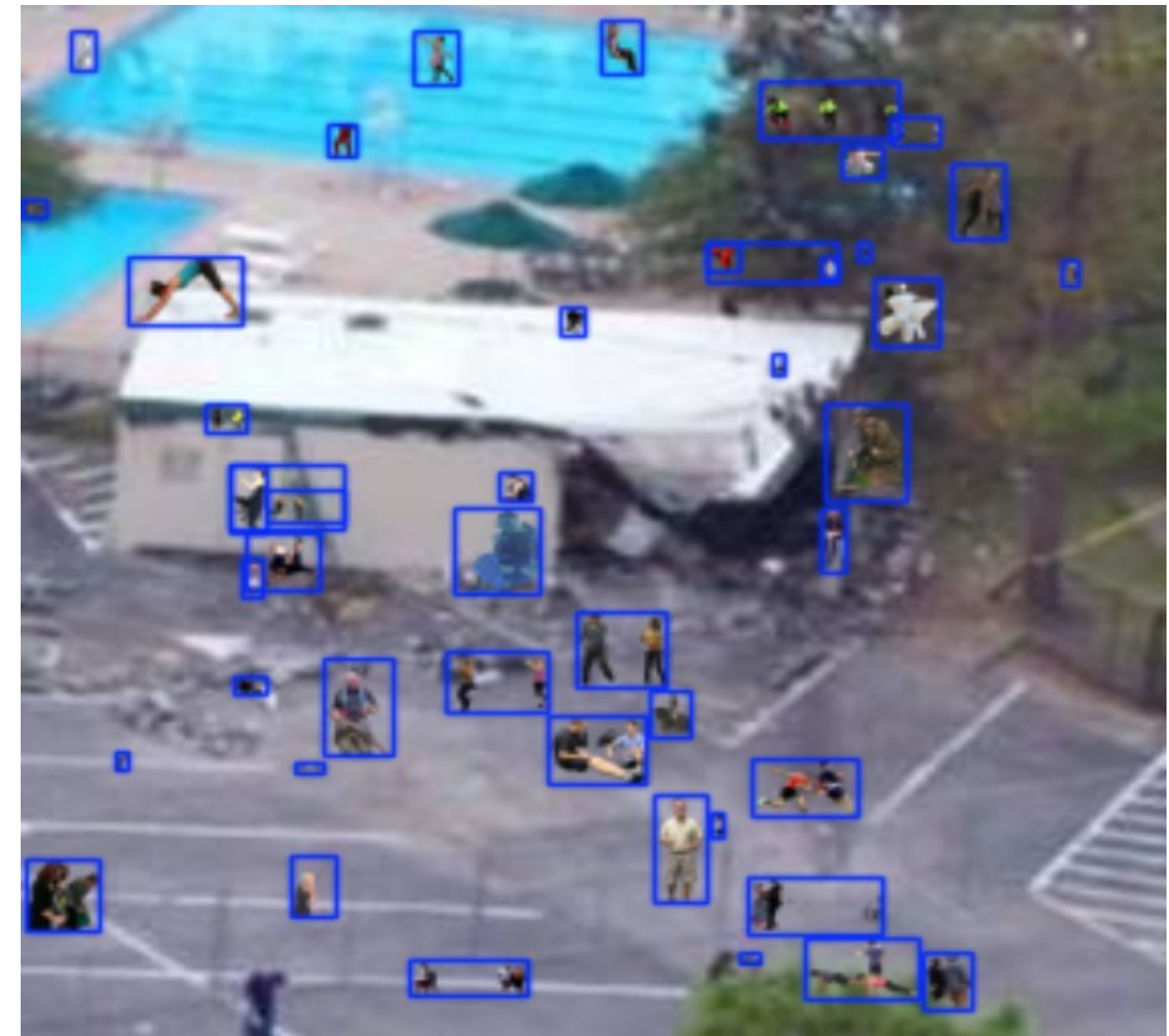


Fig. - 03: Human Detection in disaster Affected Scenario from C2A Dataset[1]



Related Works

Related Works

TABLE I — DETECTOR FAMILIES AND THEIR TRADE-OFFS

Family	Representative works	Key idea	Trade-off for our task
Two-stage	R-CNN (2014), Faster R-CNN (2015), Cascade R-CNN (2018)	propose regions, then classify	accurate but too slow/heavy for airborne use
One-stage	SSD (2016), RetinaNet + focal loss (2017), YOLO v1→v11 (2016–2024)	predict boxes in one pass	real-time, compact — our family
Transformer	DETR (2020), RT-DETR (2024)	set prediction, no NMS/anchors	competitive speed only recently; heavier

- *We work in the YOLO family because speed and model size decide airborne deployability*

Related Works(cont'd.)

TABLE II — TECHNIQUES FOR SMALL/AERIAL TARGETS

Approach	Representative works	How it helps tiny targets	Limitation in prior work
Higher-resolution scales	FPN (2017), TPH-YOLOv5 (2021), P2/stride-4 heads	detection grid fine enough for few-px targets	usually reported alone, on generic aerial data
Feature attention	SE (2018), ECA (2020), CBAM (2018)	re-weights features so faint targets survive clutter	choice asserted, rarely measured on-task
Inference-time slicing	SAHI (2022)	overlapping crops enlarge small targets	latency cost; no architecture change
Benchmarks	VisDrone (2021)	makes small objects the dominant difficulty	generic scenes, not disaster imagery

Research Gap

TABLE IV — POSITION OF THIS THESIS RELATIVE TO PRIOR WORK

Aspect	Prior work	This thesis
Small-object techniques	reported individually	compared in one controlled ablation
Attention choice	asserted or single option	CBAM vs ECA measured on the task
State-space neck	proposed as a backbone, generic data	tested in the neck vs identical baseline; reported honestly
Target domain	mostly generic aerial imagery	disaster/SAR imagery (C2A), very small humans

Problem Statement

- Given aerial image $I \rightarrow$ output boxes (x, y, w, h, c) , one per visible person.
- Correct when $\text{IoU} \geq 0.5$
- **Domain asymmetry**: a missed person may never get a second chance; a false alarm costs seconds of review.

$$\text{IoU}(\hat{B}, B) = \frac{\text{area}(\hat{B} \cap B)}{\text{area}(\hat{B} \cup B)} \geq 0.5$$

$$F_2 = \frac{5PR}{4P + R}$$

recall weighted 4× over precision — our operational metric.



Fig. - 04: One C2A test scene — 156 annotated people

Objectives

- I. Build a compact detector via an **additive ablation** — CBAM, P2 scale, SSM neck introduced **one at a time** under one frozen protocol.
- II. Identify the configuration best serving **very-tiny recall** within a deployable budget; show it performs near published SOTA.
- III. Honestly test the claim that state-space modules help detection.
- IV. Raise recall further at inference time without retraining (SAHI, TTA).

Novelty:

- CBAM as substitution
- P2 quantified, not assumed
- Single-variable ablation on C2A

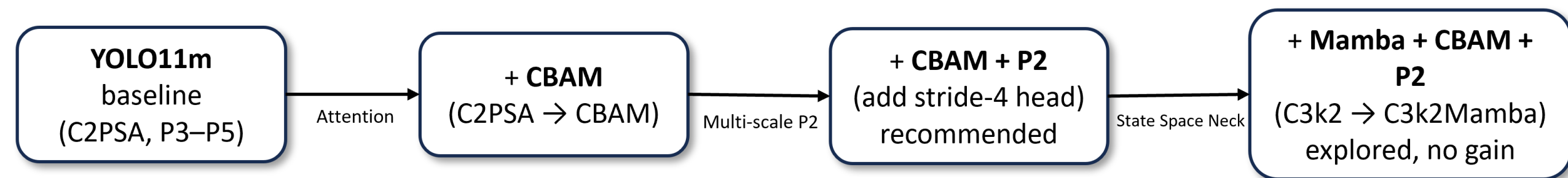


Fig. - 05 : The additive ablation chain — one change per configuration



Methodology

Methodology

Proposed MSA-YOLO Architecture:

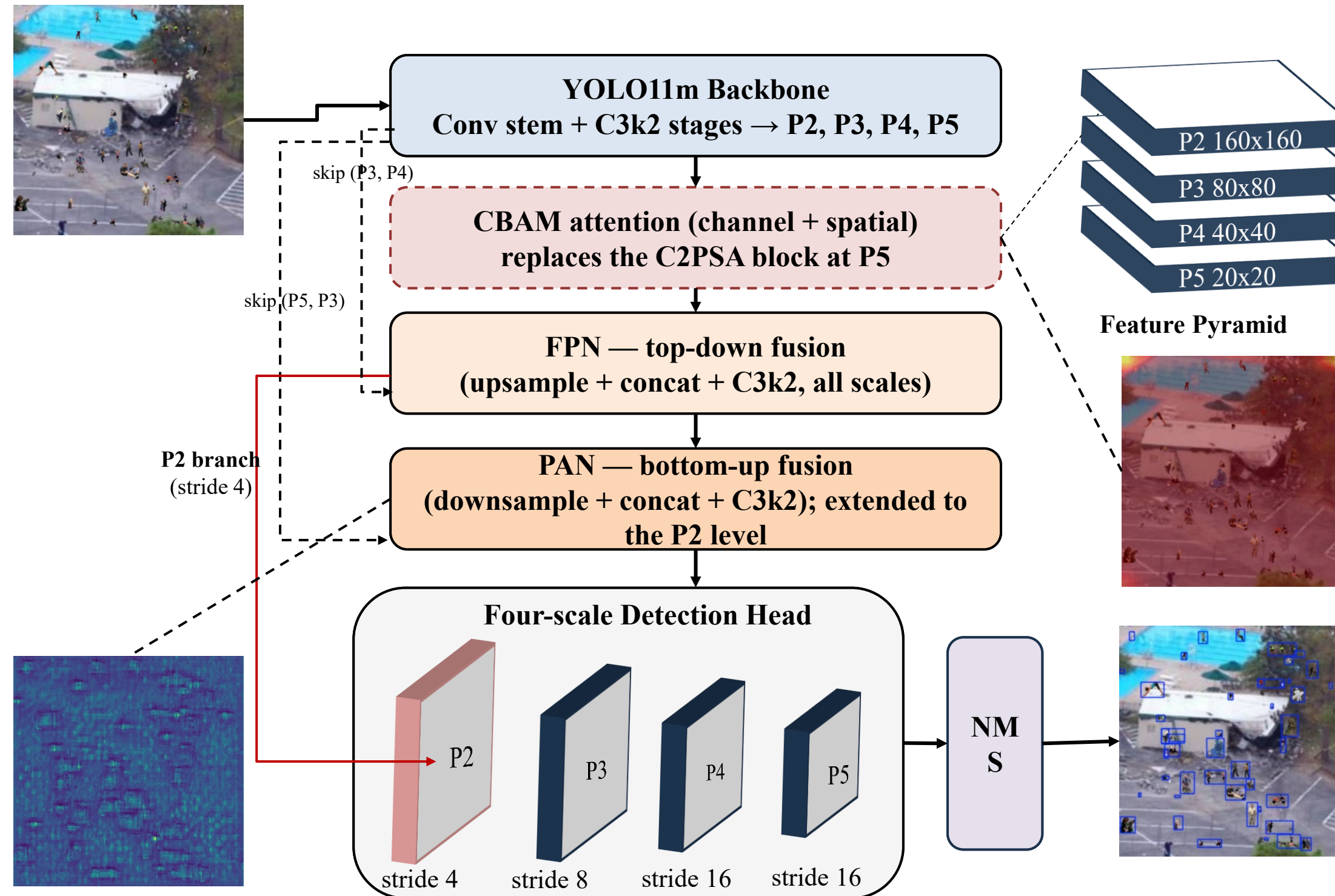
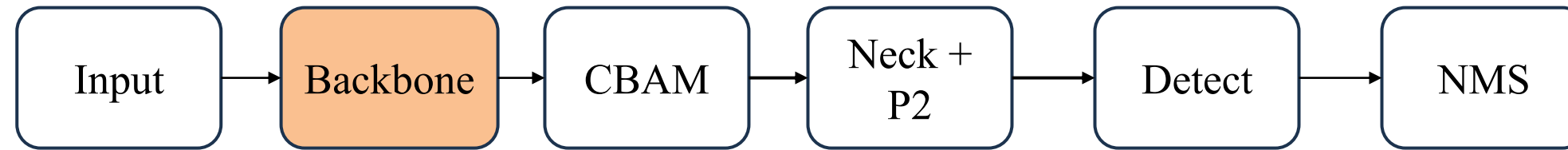


Fig. - 15 : The proposed detection pipeline

Methodology(cont'd)

Backbone:



$$P_k = \left(\frac{640}{2^k}\right)^2 \rightarrow P2\ 160^2 \cdot P3\ 80^2 \cdot P4\ 40^2 \cdot P5\ 20^2$$

- C3k2 stages extract features at strides 2–32; **layers 2/4/6 kept as skip sources.**
- Layer 2 ($160^2 \times 256$) becomes the **P2 skip** later.
- **Find:** backbone untouched — that is what keeps the ablation clean.

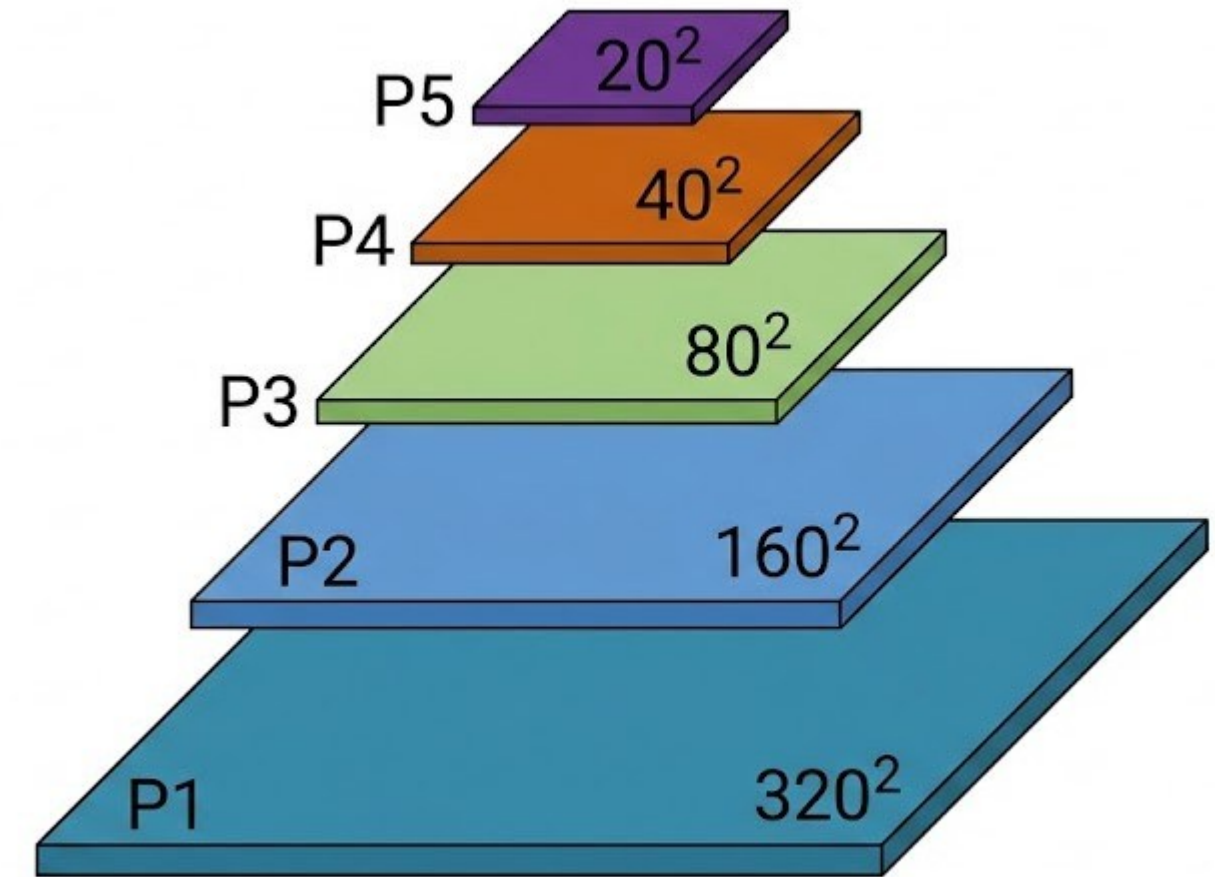
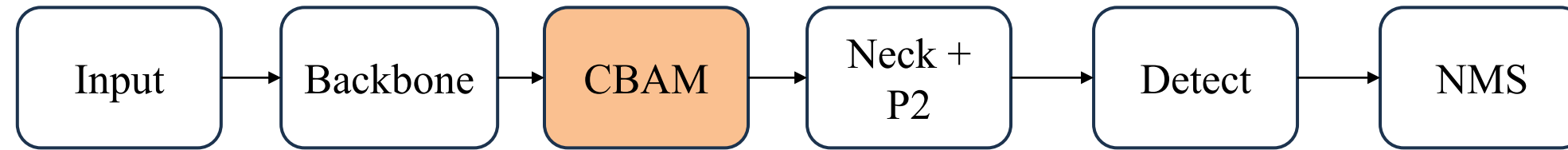


Fig. - 08 : backbone Feature Pyramid

Methodology(cont'd)

CBAM ATTENTION:



$$M_{c(F)} = \sigma \left(MLP(AvgPool(F)) + MLP(MaxPool(F)) \right)$$

$$M_{s(F')} = \sigma(f^{7 \times 7}[AvgPool(F'); MaxPool(F')])$$

$$F' = M_c \otimes F, F'' = M_s \otimes F'$$

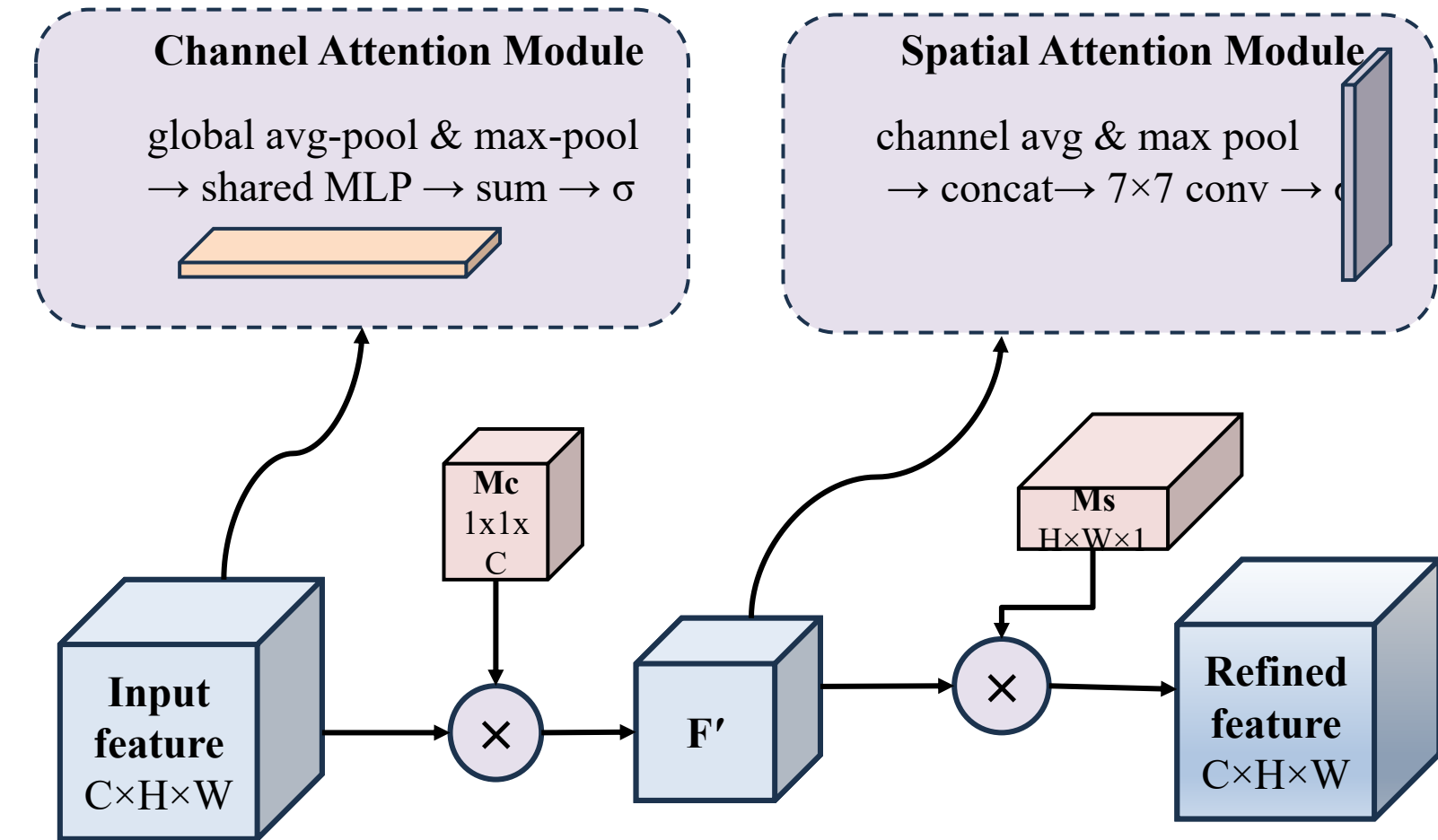
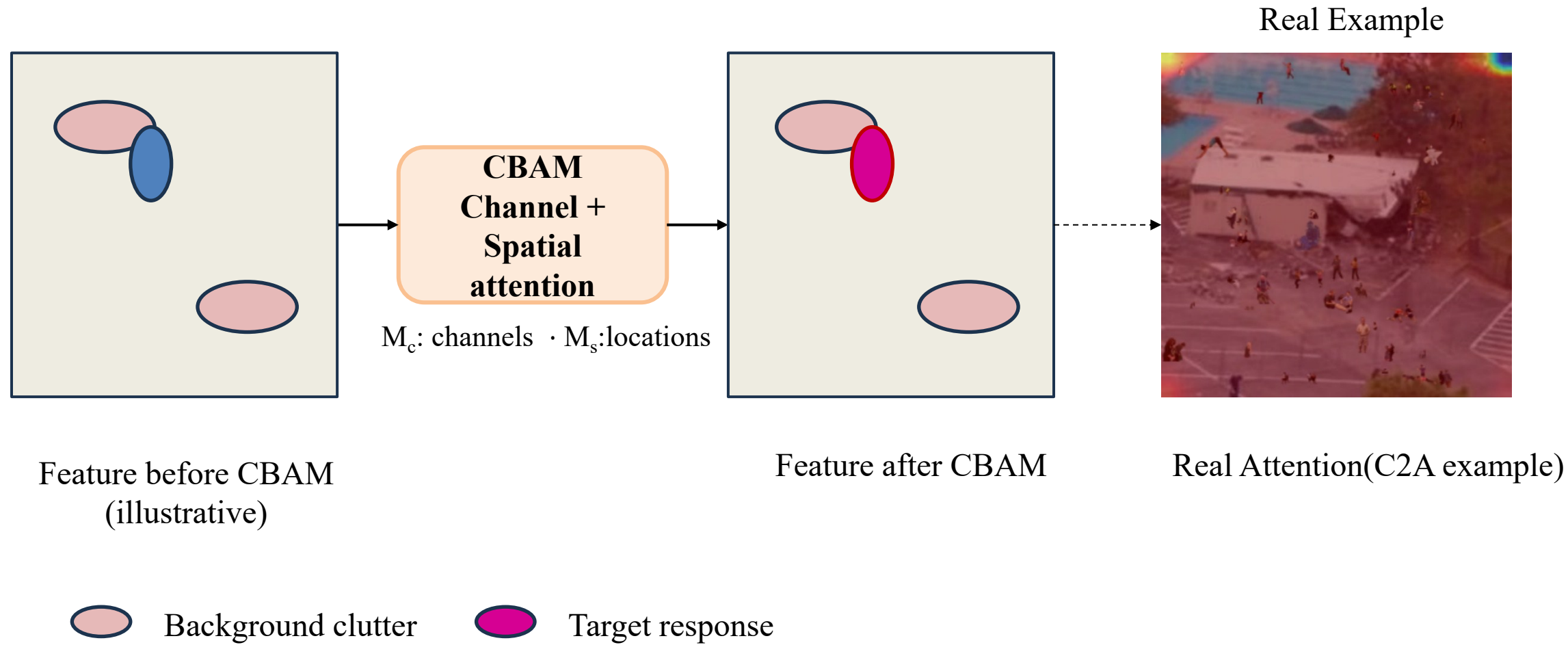


Fig. - 09 : The CBAM module — channel attention, then spatial attention

- **Break:** remove the baseline's TWO heavy C2PSA self-attention blocks.
- **Integrate:** ONE CBAM (r=16, 7×7 kernel) at layer 10 — re-weights the 512-ch P5 features.
- **Find:** ≈ 1 M fewer parameters than baseline, lowest latency of all four configs (13.5 ms), F_2 held/improved — attention at negative cost.

CBAM IN ACTION:



(a) **mechanism-** illustrative

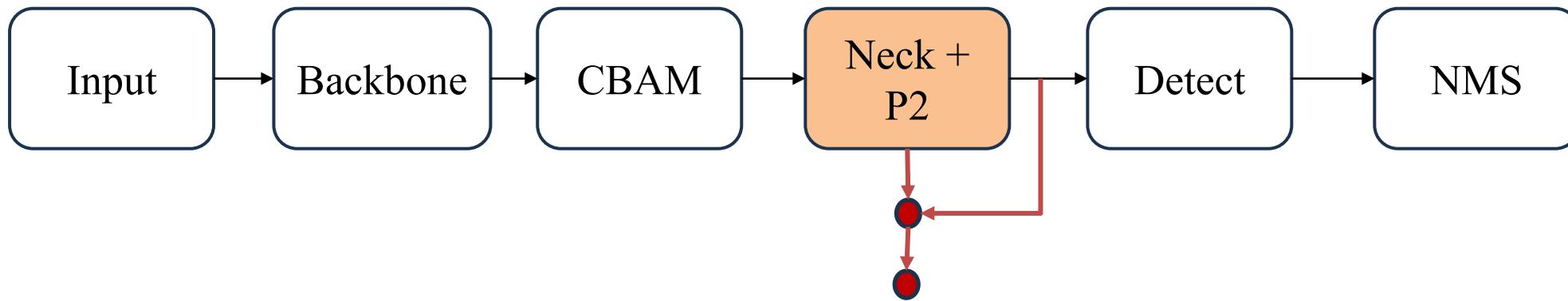
(b) **Real spatial attention** (warm = attended)

Note: *Map is 20×20 upscaled — blocky is correct; CBAM sits at stride 32*

Fig. - 10 : Clutter suppressed, target amplified — the real attention concentrates on human-shaped regions, not rubble

Methodology(cont'd)

P2 Detection SCALE:



$$c_{d(s)} = \frac{s}{d} \rightarrow 12 - px \text{ person} = 3 \text{ cells at stride 4}$$

each P2 cell covers 4×4 px $\rightarrow$ resolves targets down to ≈ 4 px

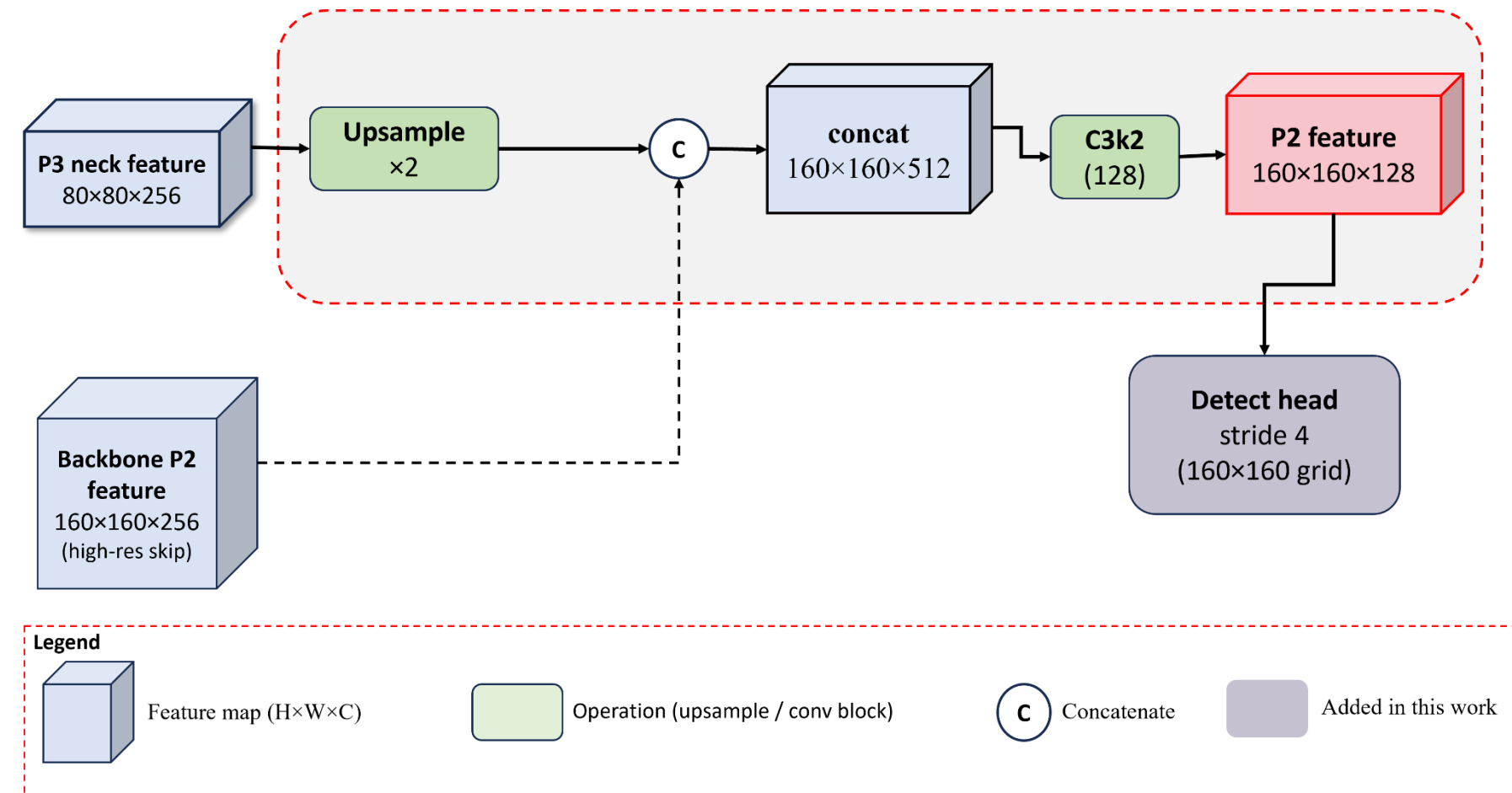
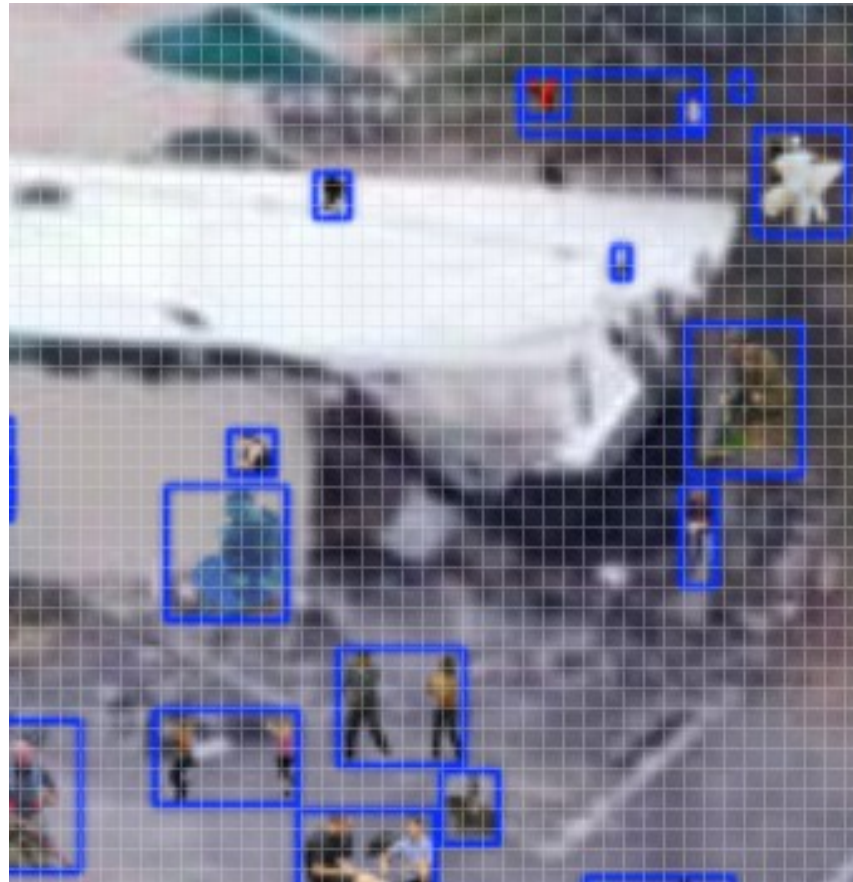


Fig. - 11 : Construction of the P2 branch

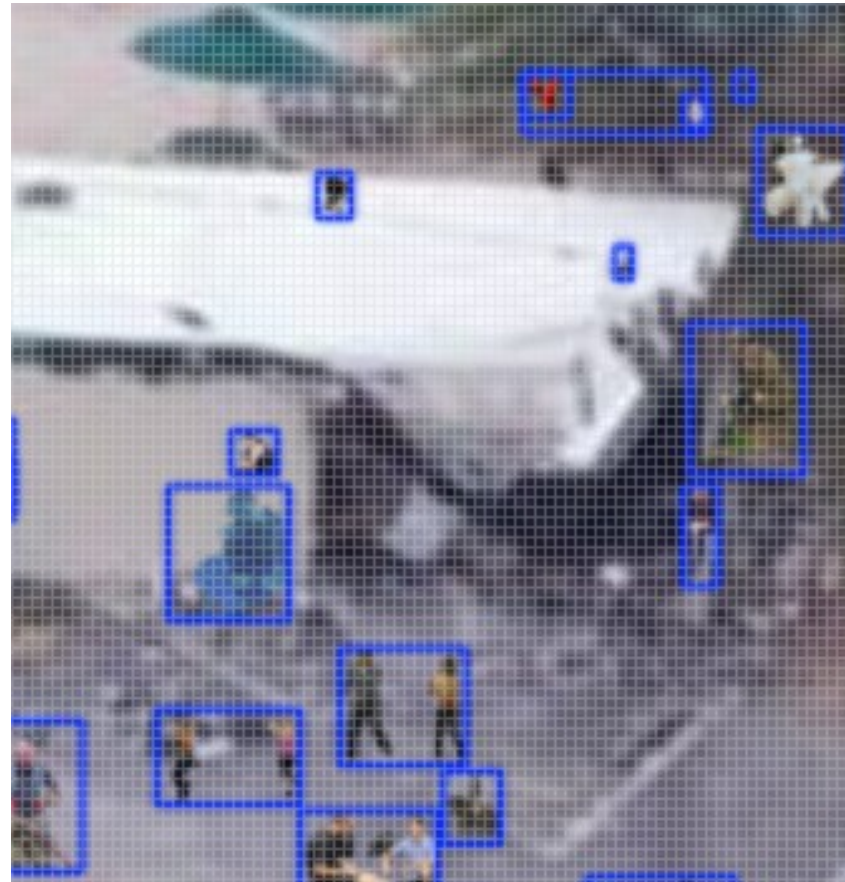
- **Integrate:** top-down pass one level further - upsample fused P3 ($80^2 \times 256$) $\rightarrow$ concat layer-2 skip ($160^2 \times 256$) $\rightarrow$ C3k2 $\rightarrow$ $160^2 \times 128$ P2 feature, stride-4 branch (neck layers 17–19).
- **Break:** nothing — P3–P5 untouched; the branch is purely additive.
- **Find:** the principal driver: very-tiny recall **0.743** $\rightarrow$ **0.757**, AP₅₀ **0.843** $\rightarrow$ **0.853** — for ~ 0.5 M params, +1 ms.

Methodology(cont'd)

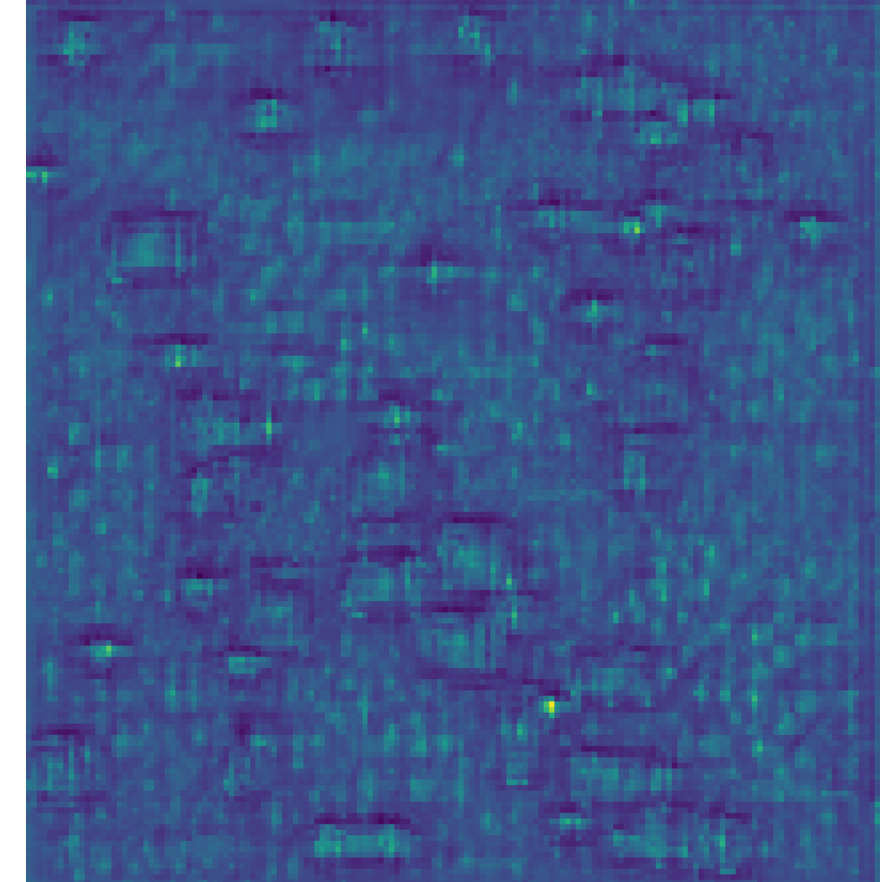
P2 IN ACTION:



(a) stride-8 (P3) grid



(b) stride-4 (P2) grid



(c) P2 feature
response

Fig. - 12 : Stride 8 assigns several neighbours to one cell; stride 4 separates them — the spatial mechanism behind the very-tiny-recall gain

Methodology(cont'd)

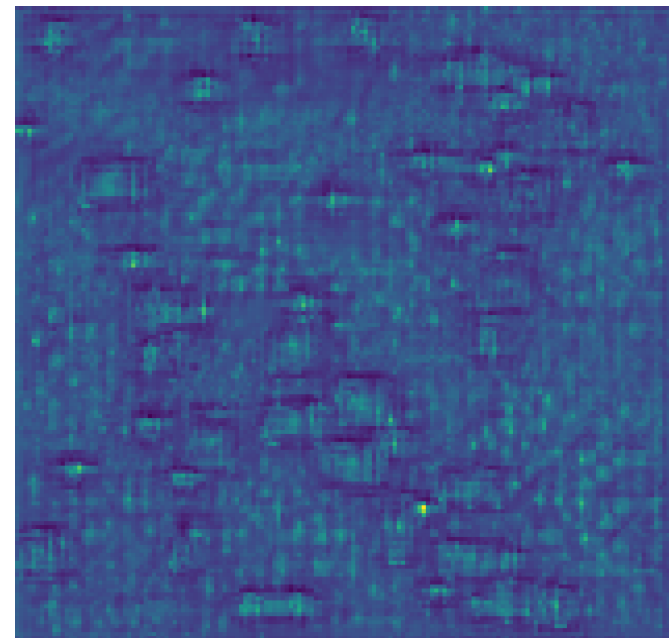
END-TO-END Trace:



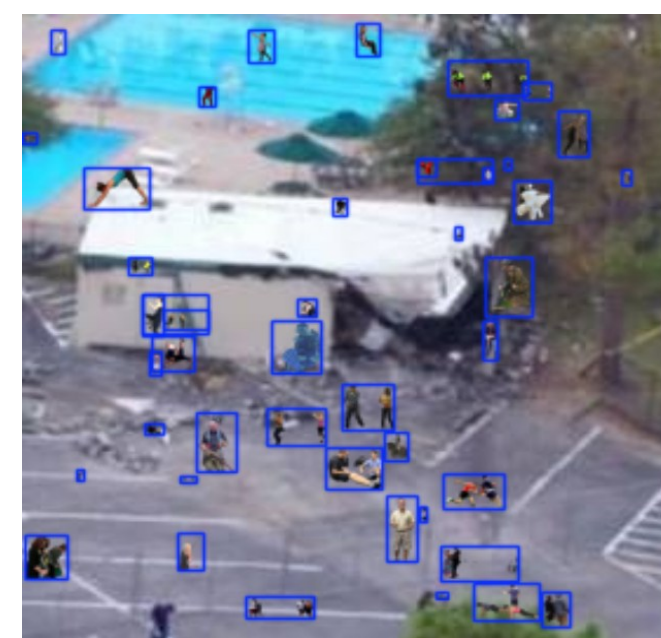
(a) input



(b) CBAM spatial attention



(c) P2 stride-4 response



(d) detections

Fig. - 15 : Every intermediate read from the trained model's activations — the pipeline does what the diagrams claim

Methodology(cont'd)

Inference-Time Enhancement: SAHI

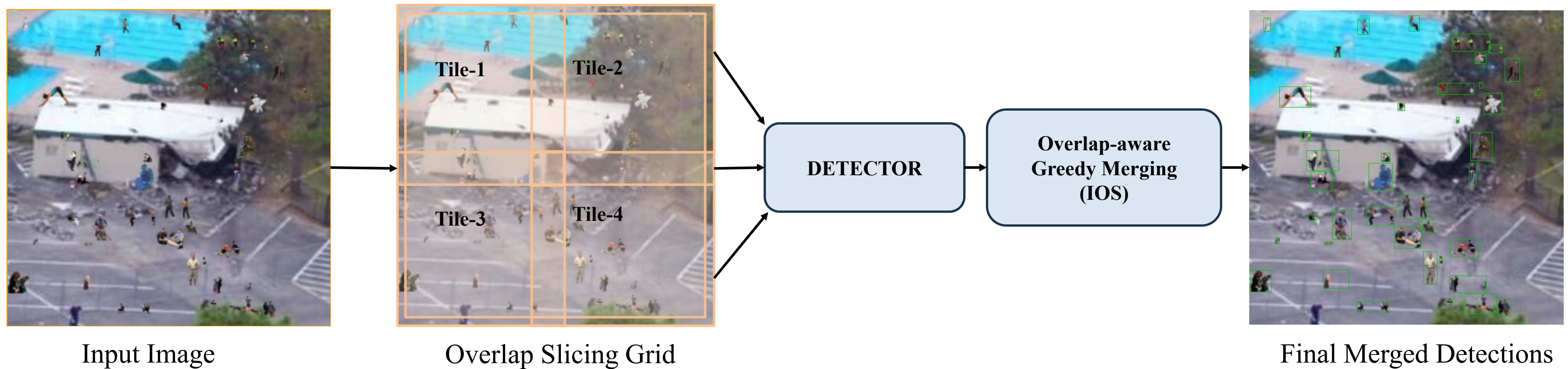


Fig. - 16 : SAHI — slice into overlapping tiles, detect per tile, merge

$$IOS(a, b) = \frac{area(a \cap b)}{\min(area(a), area(b))} \geq 0.5$$

. greedy non-maximum merging; suits boxes cut at tile borders

- **Integrate at inference only:** tiles 256–640 px, 25–30 % overlap + one full-image pass; weights untouched.
- **Find (preview):** 256-px slices → very-tiny recall 0.758 → **0.788** (+3.0 pt) — but **162 ms/image**.

Methodology(cont'd)

Inference-Time Enhancement: TTA

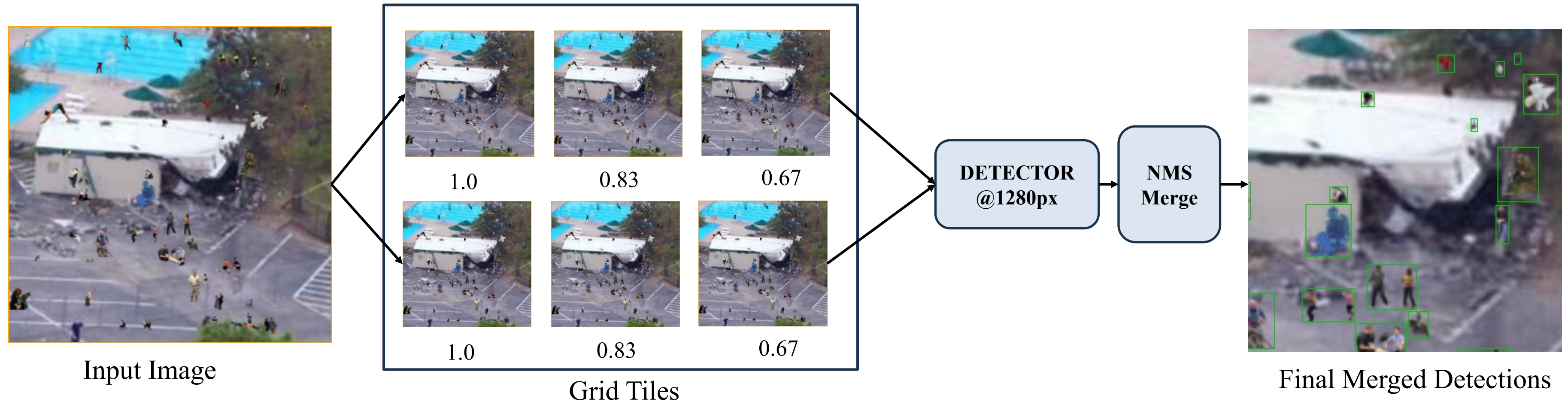


Fig. - 17 : TTA — three scales $\times$ horizontal flip at 1280 px, merged by NMS

$\{1.0, 0.83, 0.67\} \times \{id, hflip\}$
at 1280 px $\rightarrow$ NMS

Fully-convolutional $\Rightarrow$ can run at $2\times$ training resolution

- **Find (preview):** very-tiny recall $0.758 \rightarrow 0.850$ (+9.2 pt), $F_2 \rightarrow 0.854$, at only 60 ms — beats every SAHI setting. Collapses beyond $2\times$ training resolution.



Implementation & Results

Results & Findings

Experimental Setup:

Hardware/Software:

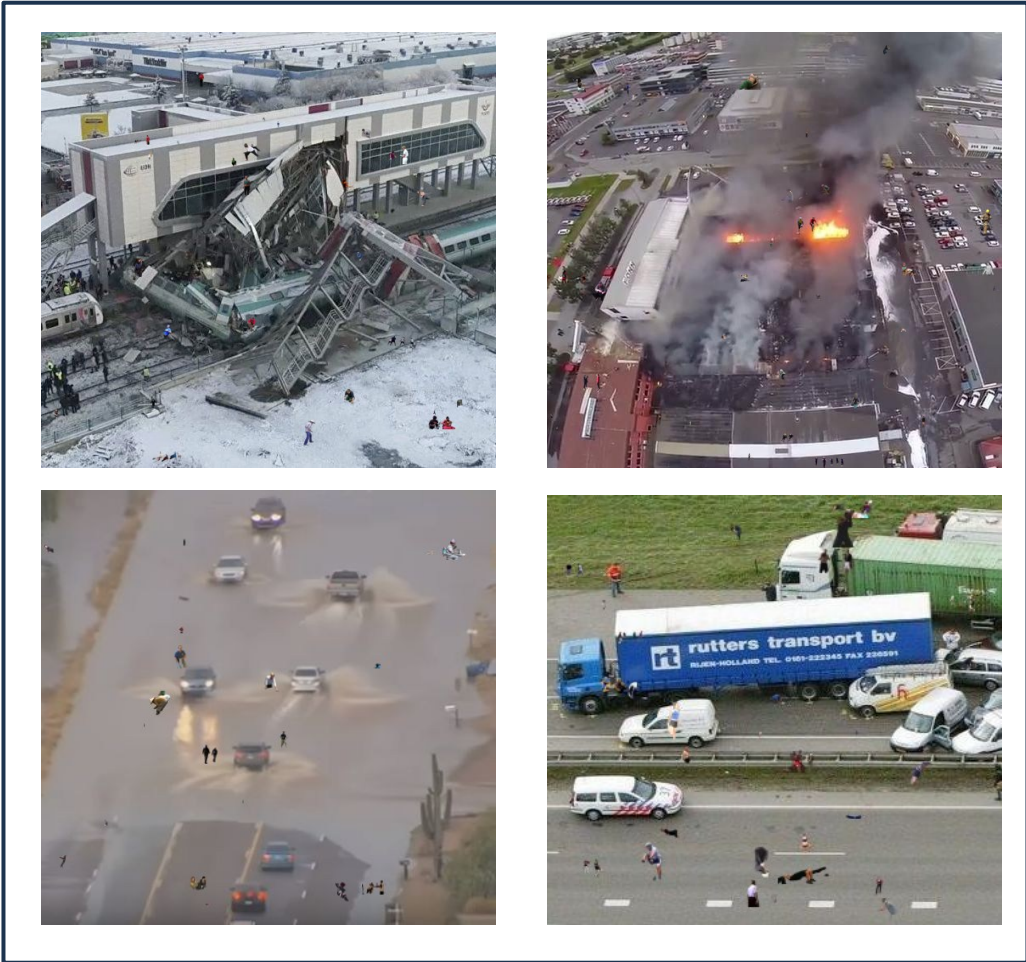
- RTX 4070 Ti SUPER 16 GB
- i7-14700K · 128 GB RAM
- Windows 11
- Python 3.11.9
- PyTorch 2.12.0 (CUDA 12.6)
- Ultralytics 8.4.56
- single GPU
- AdamW pinned because the framework's default SGD diverged on the P2 architecture (~epoch 50); AdamW trained all four stably.

TABLE VI — TRAINING CONFIGURATION (IDENTICAL FOR ALL FOUR MODELS)

Parameter	Setting
Resolution	640x640 letterbox
Optimizer	AdamW lr0 0.001-> cosine
Epochs	≤ 300 epochs, patience 50 (fitness) + 40 (F ₂)
Batch	batch 16 (P2 models: physical 8, effective 16 grad-accum)
AMP	Automatic Mixed Precision
Seed	Seed 0

Results & Findings(Cont'd.)

Dataset: C2A



- **C2A** = AIDER disaster backgrounds + U²-Net-segmented human poses (semi-synthetic); **10,215 images**, **~360k instances**, 4 scene types, 20–40 people/image.
- Frozen split (MD5-verified): **6,129 train / 2,043 val / 2,043 test** — identical for every configuration.
- Test geometry: **99.6 % < 32 px**; > 1/3 below 8 px.
- composited origin $\Rightarrow$ transfer to fully real imagery is future work.

	Train	Validation	Test
Images	6,129	2,043	2,043

Fig. - 18 : Dataset Sample and Instance table

Results & Findings(Cont'd.)

THE ADDITIVE ABLATION:

TABLE VII — ADDITIVE ABLATION ON THE C2A TEST SET (COCO

Model	Params (M)	GFLOPs	AP	PROTOCOL)		F ₁	F ₂	Lat. (ms)
				AP ₅₀	AR ₁₀₀			
YOLO11m baseline	20.03	67.7	0.615	0.843	0.691	0.850	0.840	13.7
+ CBAM	19.08	66.9	0.616	0.847	0.692	0.850	0.841	13.5
+ CBAM + P2 (MSA- YOLO)	19.57	86.7	0.615	0.853	0.703	0.848	0.844	14.6
+ Mamba + CBAM + P2	22.01	98.4	0.614	0.852	0.704	0.846	0.844	41.1

- ① P2 is the effective change (+1.0 AP₅₀ pt, +1.2 AR pt for 0.5 M / 1 ms)
- ② CBAM near-free (−1 M params, fastest)
- ③ Mamba: +2.4 M, 2.8× latency.

Results & Findings(Cont'd.)

Component Attribution:

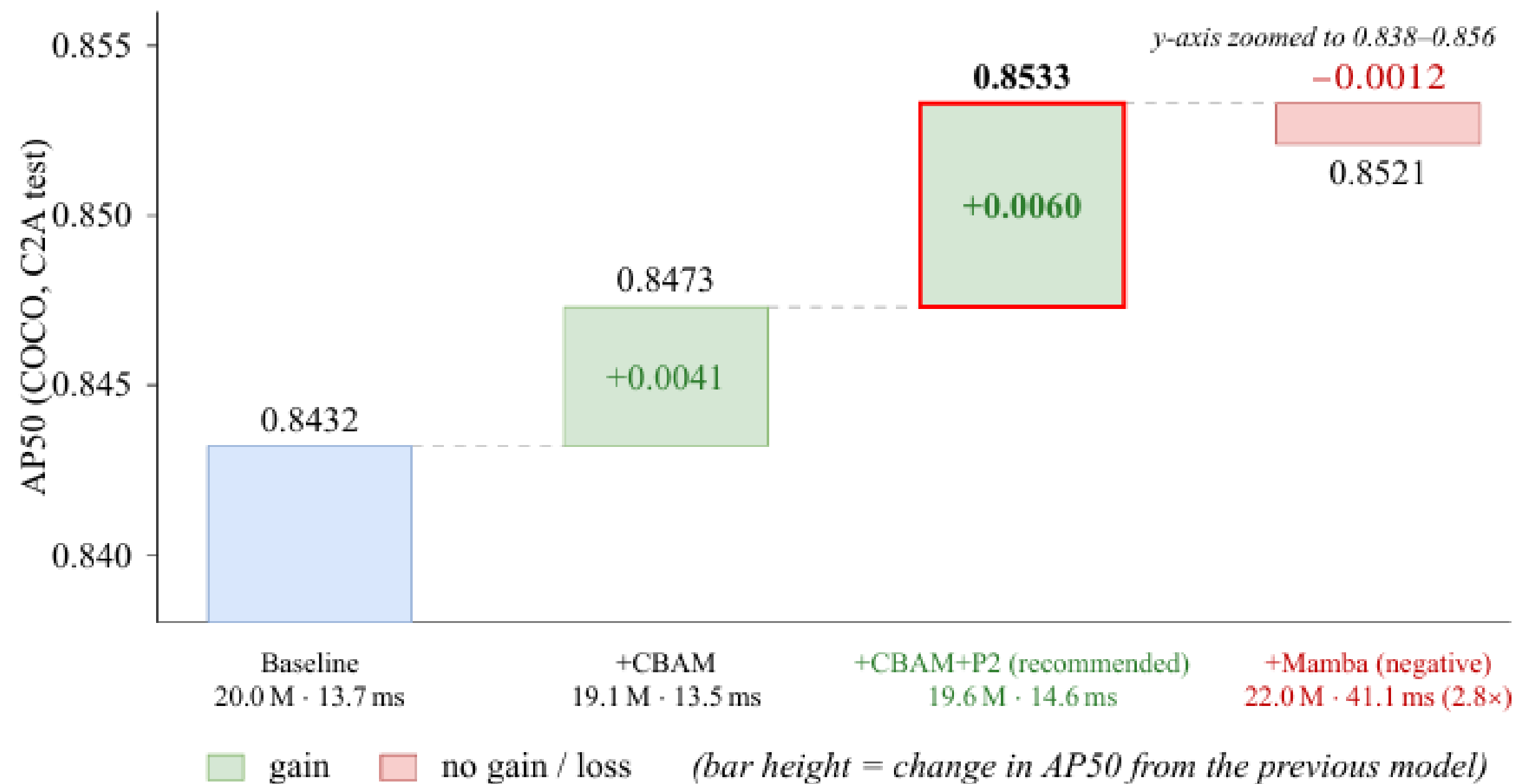


Fig. - 19 : Per-component contribution to AP₅₀ (y-axis zoomed 0.838–0.856); params · latency under each column

Note: Every gain attributed to one change; the Mamba null is displayed, not hidden.

Results & Findings(cont'd.)

HYPOTHESIS TEST: PER-SIZE RECALL

TABLE VIII — PER-SIZE RECALL ON THE C2A TEST SET

Model	very-tiny (<8 px)	tiny (8–16)	small (16–32)	medium (32–96)
GT instances	25,072	20,520	26,614	317
YOLO11m baseline	0.743	0.869	0.894	0.909
+ CBAM	0.746	0.873	0.895	0.912
+ CBAM + P2	0.757	0.865	0.886	0.808
+ Mamba + CBAM + P2	0.757	0.870	0.887	0.811

- Medium-band drop is noisy: only **317** of **~72,500** instances.
- Mamba tracks CBAM+P2; scan diagnostics prove the blocks ran (fwd/rev cosine distance **0.836**)

Results & Findings(Cont'd.)

Component Attribution:

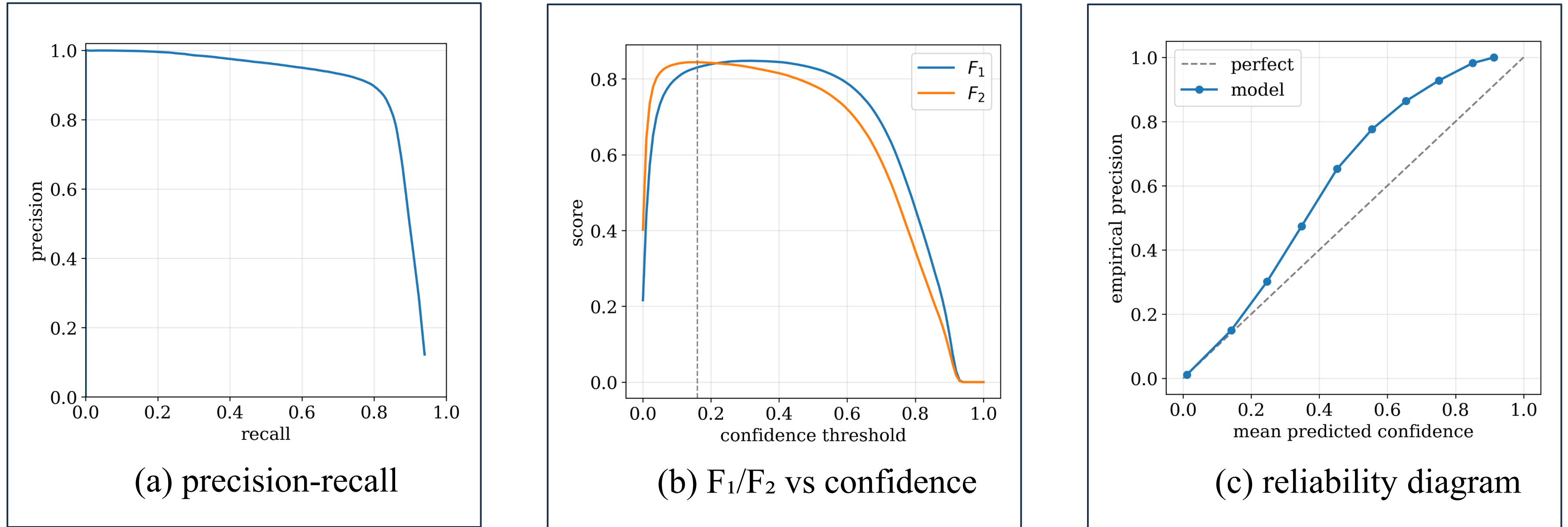


Fig. - 20 : MSA-YOLO operating behaviour on the C2A test set

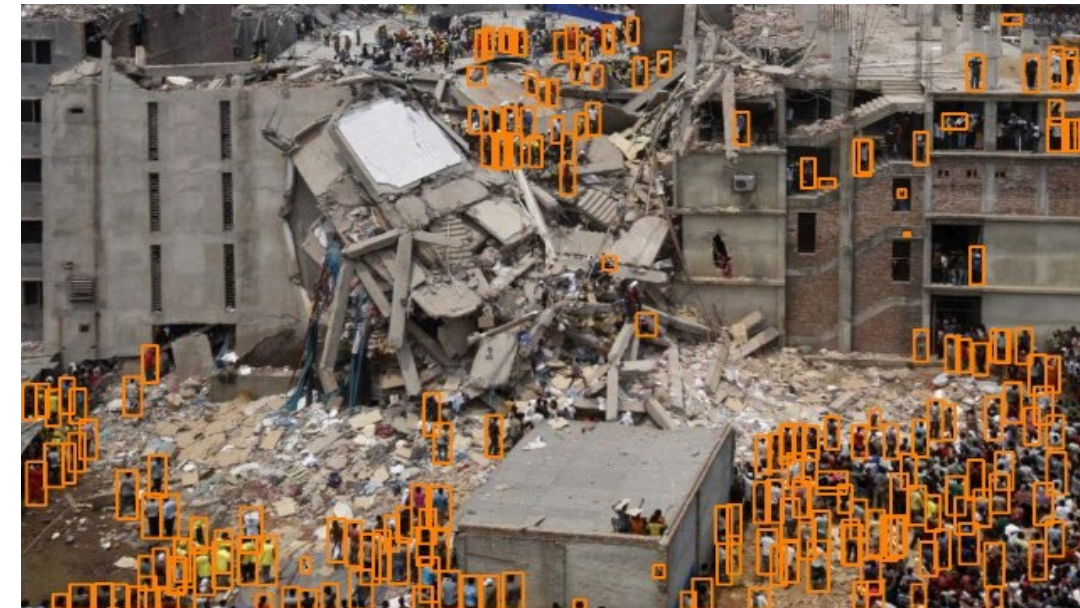
- F_2 -optimal threshold **0.16–0.20 across all configs** $\Rightarrow$ no per-model tuning; MSA-YOLO deploys at **0.16**.
- **ECE $\approx$ 0.021** $\Rightarrow$ reported 0.8 confidence $\approx$ 0.8 empirical precision — operators can trust the scores for triage.

Results & Findings(cont'd.)

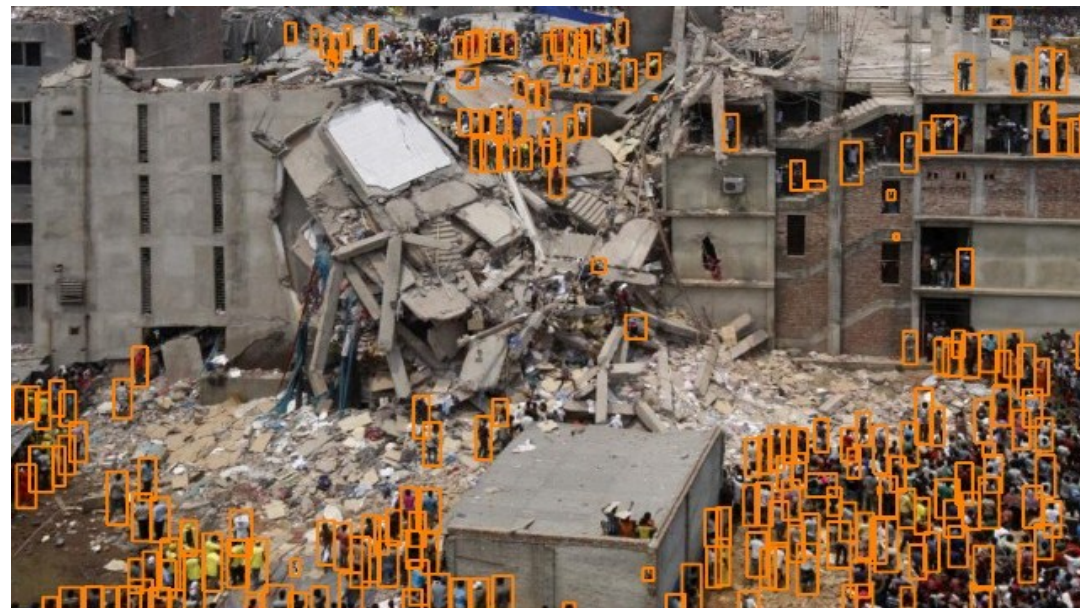
QUALITATIVE ANALYSIS:



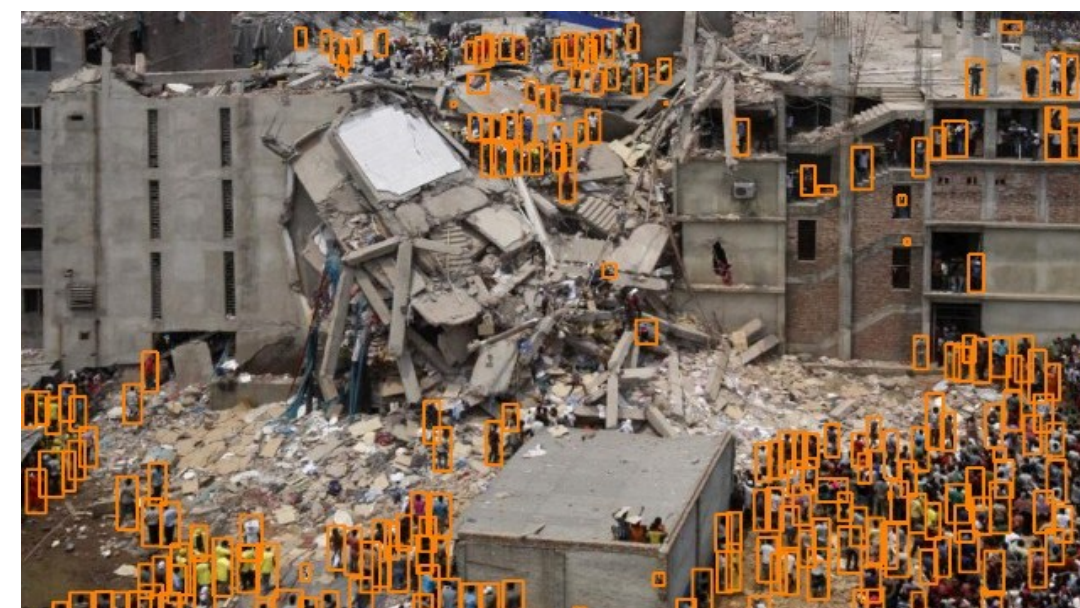
(a) Ground truth — 156 people



(b) MSA-YOLO @640 px — 209 boxes



(c) + SAHI 256 — 217



(d) + SAHI+TTA — 217

Fig. - 21 : Dense collapsed-building scene — sliced/augmented modes add detections along the rubble line

Results & Findings(cont'd.)

COMPARISON WITH SOTA:

TABLE IX — PUBLISHED DETECTORS ON THE C2A TEST SET (Nihal et al., same split, no

Model	AP ₅₀	AP	Params
Faster R-CNN	0.634	0.366	~41 M
RetinaNet	0.693	0.383	~37 M
RTMDet	0.708	0.442	—
Cascade R-CNN	0.735	0.486	~69 M
DINO	0.789	0.471	~47 M
YOLOv5	0.808	0.492	—
YOLOv9-e	0.893	0.688	57.3 M
YOLO11m baseline (ours)	0.843	0.615	20.1 M
MSA-YOLO (ours)	0.853	0.615	19.6 M

- Second among ALL published detectors on this benchmark — one AP₅₀ point behind a model THREE TIMES our size.

Results & Findings(cont'd.)

Inference-Time Study: SAHI vs TTA

TABLE X - SAHI SWEEP AND TTA (MSA-YOLO, PER-IMAGE MATCHING @ IoU 0.5)

Setting	Size	R	F ₂	very-tiny R	Lat. (ms)
none (baseline)	640	0.835	0.839	0.758	15
+ SAHI	256	0.846	0.848	0.788	162
+ SAHI	512	0.837	0.842	0.763	66
+ TTA	1280	0.877	0.854	0.850	60

- TTA@1280 is the single best setting — +9.2 pt very-tiny recall, cheaper than any SAHI config; degrades beyond 2× training resolution.
- Both are optional modes — neither touches the trained weights.

Results & Findings(cont'd.)

Live Demo(on Cross Domain for testing Generalization):



Video - 01 : Drone footage Test

Sources:

- KUET Central Field shot with DJI Mini 3S(Settings: speed- 5.6-9.7m/s , Altitude- 50m/30m/10m , Distance Covered – 103m)
- News Videos – sourced from The Guardian[31] & AP News[32]

Thesis Timeline

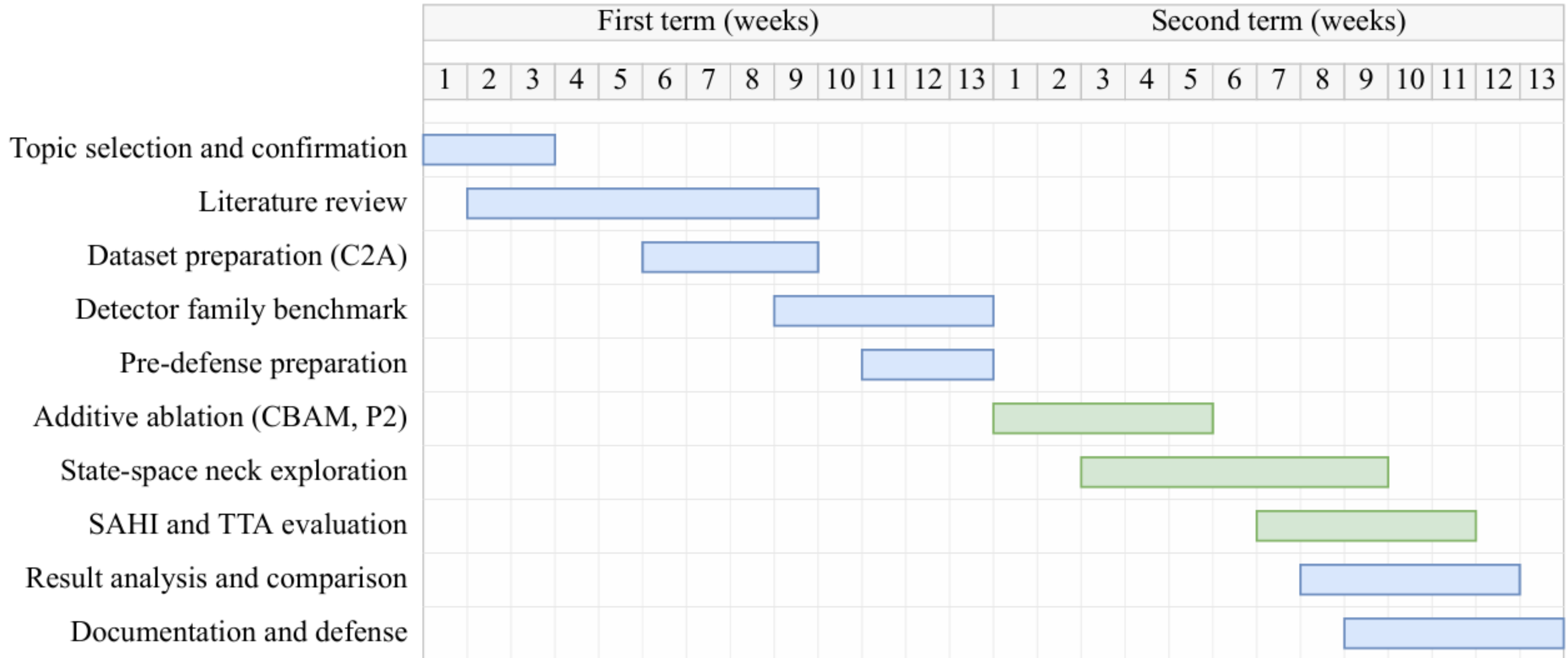


Fig-22: Thesis Timeline Gantt Chart for the Full Term

Limitations

- **Single, semi-synthetic dataset (C2A) — transfer to fully real footage untested here.**
- **Sparse medium band — only 317 instances; its estimates are noisy**
- **Edge-device optimization for UAV deployment**

Future Work

- **Server-side deployment** — drone streams to ground server; detection + flagged frames back to operator; enables TTA@1280.
- **Enhanced dataset with own drone imagery** — attacks the composited-data domain gap directly.
- **Validation on real rescue imagery** — SARD, HERIDAL, VisDrone, Okutama (zero-shot + fine-tuned).

Conclusion

- **The gain on C2A comes almost entirely from the P2 detection scale** — the resolution argument was right.
- **CBAM contributes efficiency**: attention at negative parameter cost.
- **Results validate feasibility on disaster datasets**
- **The state-space neck contributes nothing at $2.8\times$ latency** — an honest, verified null

References

- [1] R. A. Nihal, B. Yen, K. Itoyama, and K. Nakadai, "UAV-enhanced combination to application: Comprehensive analysis and benchmarking of a human detection dataset for disaster scenarios," in *Proc. Int. Conf. Pattern Recognit. (ICPR)*, 2024, doi: 10.48550/arXiv.2408.04922.
- [2] G. Jocher and J. Qiu, "Ultralytics YOLO11," 2024. [Online]. Available: <https://github.com/ultralytics/ultralytics>
- [3] X. Zhu, S. Lyu, X. Wang, and Q. Zhao, "TPH-YOLOv5: Improved YOLOv5 based on transformer prediction head for object detection on drone-captured scenarios," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, 2021, pp. 2778–2788.
- [4] AGIIndia, "Life-saving insights in disaster zones with drones," Apr. 2024. [Online]. Available: <https://agiindia.com/life-saving-insights-in-disaster-zones-with-drones/> [Accessed: Dec. 14, 2025].

References (Cont'd...)

- [5] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2014, pp. 580–587.
- [6] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, 2017.
- [7] Z. Cai and N. Vasconcelos, "Cascade R-CNN: Delving into high quality object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2018, pp. 6154–6162.
- [8] W. Liu et al., "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2016, pp. 21–37.
- [9] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2017, pp. 2980–2988.
- [10] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 779–788.

References (Cont'd...)

- [11] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in Proc. Eur. Conf. Comput. Vis. (ECCV), 2020, pp. 213–229.
- [12] Y. Zhao et al., "DETRs beat YOLOs on real-time object detection," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2024.
- [13] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), 2017, pp. 2117–2125.
- [14] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2018, pp. 7132–7141.
- [15] Q. Wang, B. Wu, P. Zhu, P. Li, W. Zuo, and Q. Hu, "ECA-Net: Efficient channel attention for deep convolutional neural networks," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2020, pp. 11534–11542.
- [16] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in Proc. Eur. Conf. Comput. Vis. (ECCV), 2018, pp. 3–19.

References (Cont'd...)

- [31] Guardian News, "Drone footage shows extent of flood damage to Greek island," YouTube, Apr. 4, 2025. [Online Video]. Available: <https://www.youtube.com/watch?v=yABNNdKFhkU>. [Accessed: Jul. 11, 2026].
- [32] Associated Press, "Drone footage shows widespread destruction in Gaza," YouTube, Jan. 22, 2025. [Online Video]. Available: <https://www.youtube.com/watch?v=P11jiFgot5o>. [Accessed: Jul. 11, 2026].



THANK YOU

const Life/Thesis = [...solved , ...unsolved]

